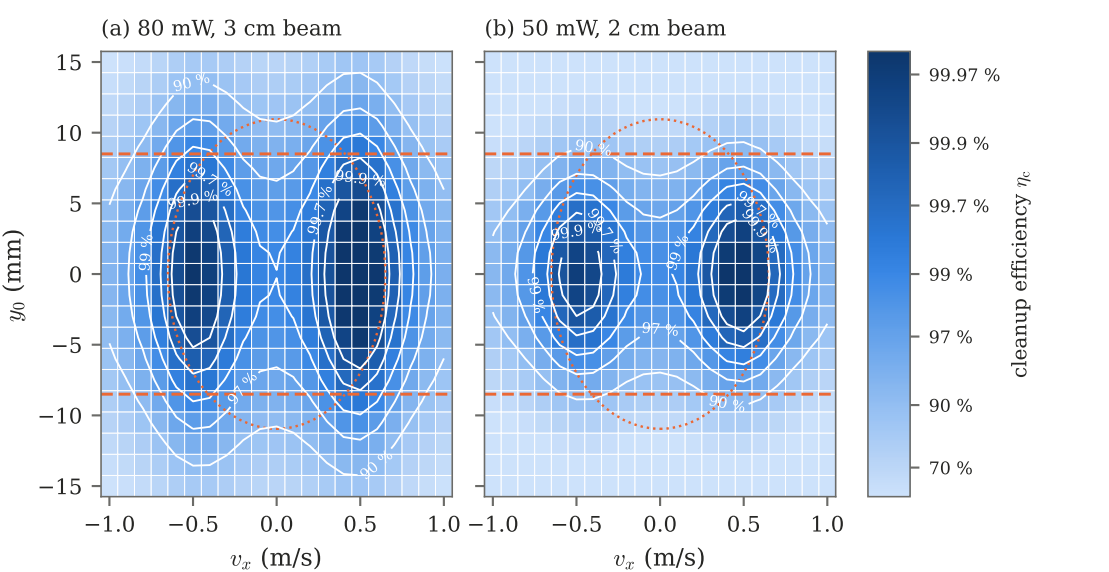


R(1) $F_1' = 5/2$, $F' = 2$; 150 V/cm
$\delta/2\pi = +19.0$ MHz, no EOM
80 mW, 3 cm high ($\sigma_y = 7.5$ mm)
start: 8 SPB-related $J = 1$ levels
on axis: 99.01 %
mean in acceptance : 98.7 %

R(1) $F_1' = 5/2$, $F' = 2$; 150 V/cm
$\delta/2\pi = +19.0$ MHz, no EOM
50 mW, 2 cm high ($\sigma_y = 5$ mm)
start: 8 SPB-related $J = 1$ levels
on axis: 98.78 %
mean in acceptance : 96.5 %

Dashed orange: detection acceptance $|y_0| \le 8.5$ mm (detection laser 1.7 cm high); the grid spans the accepted $|v_x| \le 1$ m/s. Dotted orange: 1σ of the beam distribution at CU1 (lens + 0.5 m, lens at 25 kV): $\sigma_{v_x} = 0.65$ m/s, $\sigma_y = 10.95$ mm; it weights the mean in acceptance. White lines: contours of η_c. On axis: $v_x = 0$, $y_0 = 0$. Beam heights are full 4σ widths; the power normalisation scales with the height.
13 passes, 10 % loss per pass, X/Z polarization switching at 0.5 MHz, Doppler shift $\pm k v_x$ alternating pass by pass; single trajectories at $v_z = 184$ m/s; grid 0.1 m/s × 1.5 mm, computed for $y_0 \ge 0$ and mirrored. (a) centrex-tlf 0.2.5, (b) 0.3.0 (CU1 agrees to 10^{-6}).